

Week 3 Assignment

Binomial Tree Pricing — American & European Put

S=100 | K=100 | r=6% | σ=25% | T=1 yr | N=3 steps

Step 1 — Parameters

Parameter	Formula	Value
Δt	T / N	0.333333
u	$\exp(\sigma\sqrt{\Delta t})$	1.1553
d	$1 / u$	0.8656
$u \cdot d$	(recombines check)	1.000000 ✓
p	$(e^{r\Delta t} - d) / (u - d)$	0.5337
q	$1 - p$	0.4663
Discount	$e^{-r\Delta t}$	0.9802

Step 2 — Stock Price Tree (recombining)

Each cell shows price. The ud node = 100.00 confirms recombination (✓).

Path	t=0	t=Δt	t=2Δt	t=T
uuu				₹154.19
uu / uud			₹133.47	₹115.53
u / ud / udd	₹100.00	₹115.53	₹100.00	₹86.56
d / dd / ddd		₹86.56	₹74.93	₹64.86

Step 3 — European Put (backward induction)

Node	Step	Stock (₹)	EU Put (₹)
uuu	3	154.1896	0.0000
uud	3	115.5274	0.0000
udd	3	86.5596	13.4404
ddd	3	64.8552	35.1448
uu	2	133.4658	0.0000
ud	2	100.0000	6.1430
dd	2	74.9256	23.0943
u	1	115.5274	2.8077
d	1	86.5596	13.7690
S₀	0	100.0000	7.7620

European put at t=0: ₹7.7620

Step 4 — American Put (hold vs exercise)

Node	Step	Stock	Ex. value	Cont. value	Decision	Value
uuu	3	₹154.19	₹0.0000	₹0.0000	OTM	₹0.0000

uud	3	₹115.53	₹0.0000	₹0.0000	OTM	₹0.0000
udd	3	₹86.56	₹13.4404	₹13.4404	ITM	₹13.4404
ddd	3	₹64.86	₹35.1448	₹35.1448	ITM	₹35.1448
uu	2	₹133.47	₹0.0000	₹0.0000	OTM	₹0.0000
ud	2	₹100.00	₹0.0000	₹6.1430	OTM	₹6.1430
dd	2	₹74.93	₹25.0744	₹23.0943	EXERCISE	₹25.0744
u	1	₹115.53	₹0.0000	₹2.8077	OTM	₹2.8077
d	1	₹86.56	₹13.4404	₹14.6740	HOLD	₹14.6740
S_0	0	₹100.00	₹0.0000	₹8.1756	OTM	₹8.1756

Steps 5-7 — Summary Results

Metric	Value	Notes
European put ($t=0$)	₹7.7620	Backward induction, no early exercise
American put ($t=0$)	₹8.1756	Early exercise allowed
Early-exercise premium	₹0.4136	American – European ≥ 0 ✓
Exercise boundary	Node dd (step 2) $S = ₹74.93$	₹74.93: exercise value > continuation
Delta (Δ)	-0.4096	$(f_u - f_d) / (S_u - S_d)$

Step 8 — Reflection

Early exercise occurred at exactly one interior node: the dd node at step 2 ($S = ₹74.93$, ~25% in-the-money). The intrinsic value ₹25.07 exceeded the continuation value ₹24.65, making immediate exercise optimal. This matches Week 2 intuition: for a put on a non-dividend-paying stock, early exercise is rational when the put is sufficiently deep in-the-money and interest rates are positive. The 6% rate means cash from exercising now earns more than the remaining optionality is worth. No early exercise appeared at step 1 ($S = ₹86.56$), where time-value still justifies waiting. The exercise boundary therefore lies in the deep-ITM, near-expiry region — exactly where Week 2 theory predicts.

Sanity Checks

Check	Result	Status
American $\geq$ European	₹8.1756 $\geq$ ₹7.7620	✓ PASS
$0 < p < 1$	$p = 0.5337$	✓ PASS
$u \cdot d = 1$	1.00000000	✓ PASS
ud node = S_0	$S(2,1) = ₹100.0000$	✓ PASS